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UniM: A Unified Any-to-Any Interleaved Multimodal Benchmark

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Paper Code

TL;DR

Moving beyond isolated modality pairs toward unified any-to-any interleaved multimodal intelligence, where models understand and generate arbitrarily combined multimodal content. Interleaved setting bring new challenges for truly unified multimodal systems.

Unified Understanding and Generation; Any-to-Any Multimodal

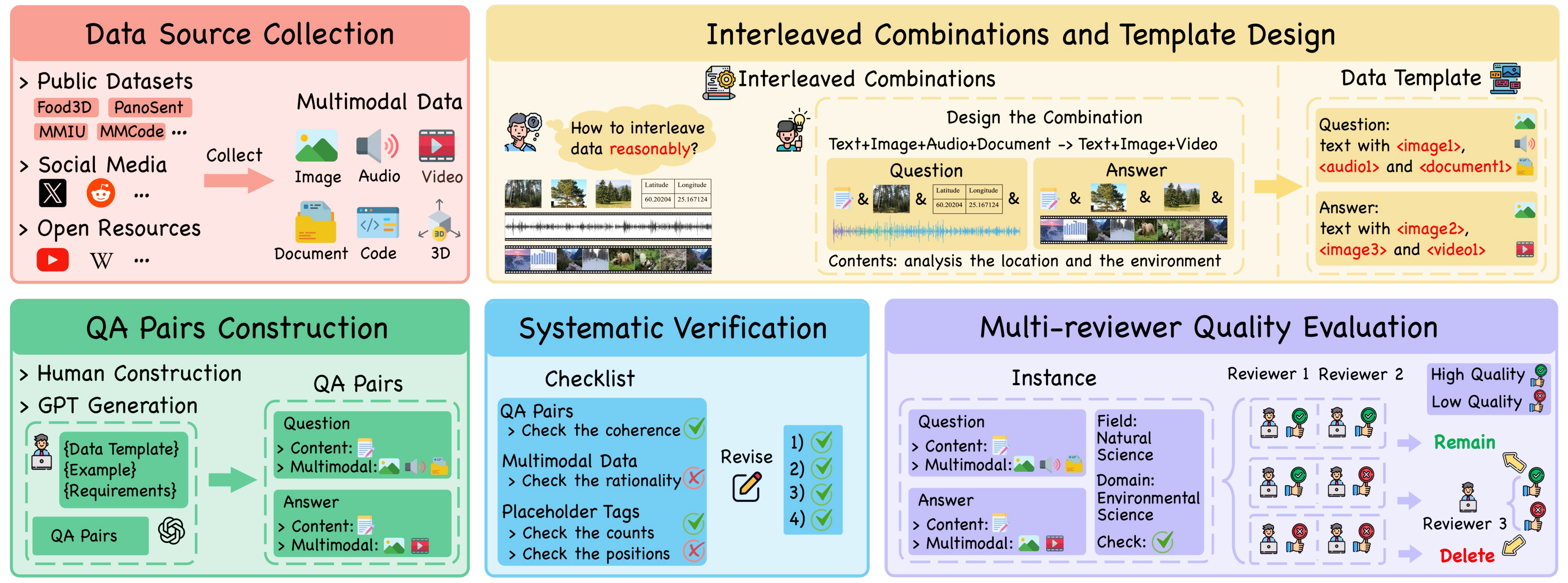
UniM: Unified A2A Interleaved Benchmark

Highlights and Challenges of UniM

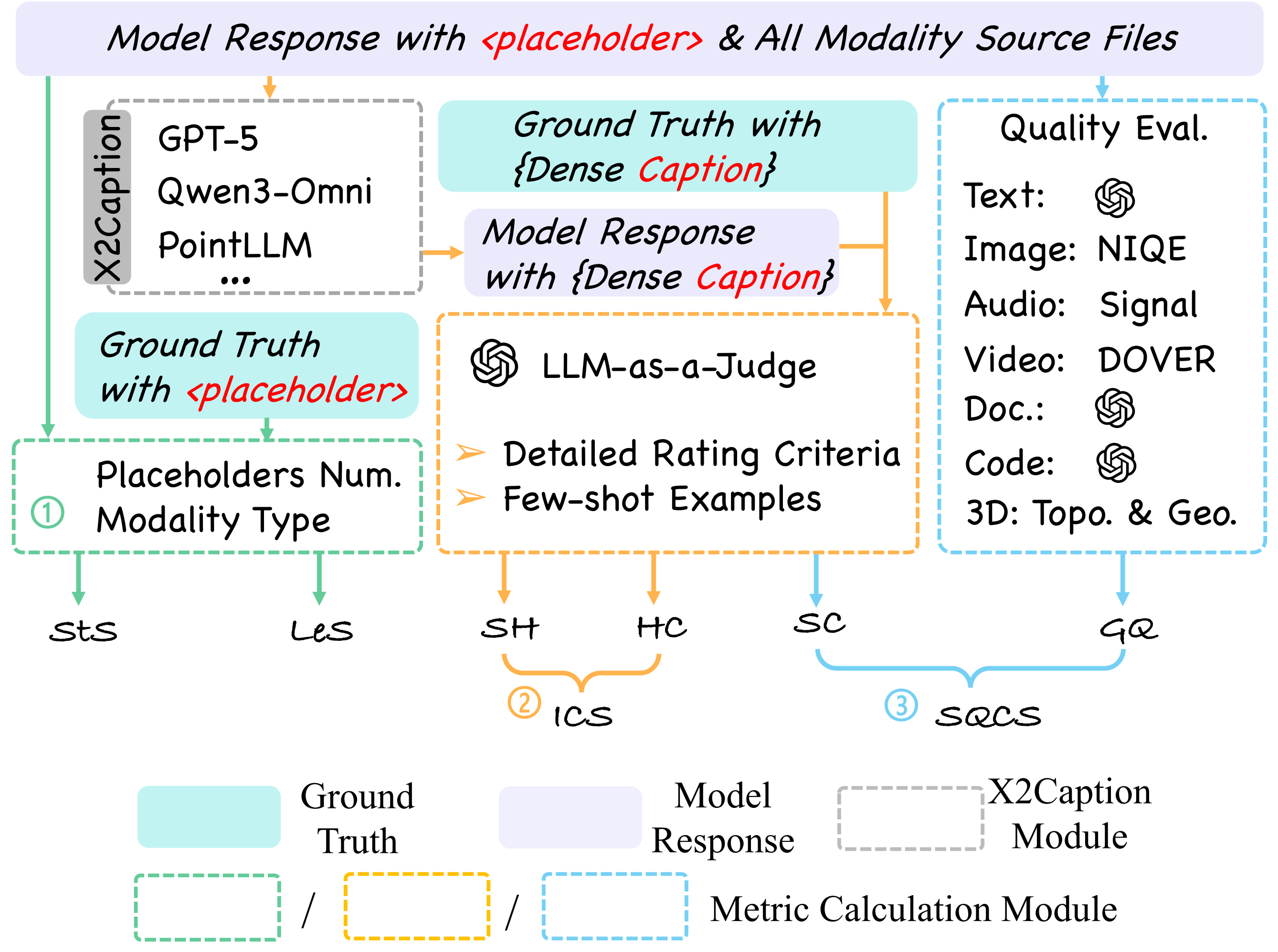
(1) Any-to-Any Interleaved Modalities. (text, image, audio, video, doc., code, 3D)
(2) Universal and Diverse Capabilities. (10 core capabilities)
(3) Multi-domain Coverage. (3 fields, 30 domains)
(4) Multiple Tasks per Instance. (multiple tasks required in the same instance)
(5) Progressive Difficulty. (3 levels, easy, medium, hard)
(6) Large Scale and High Quality. (31,026 instances)

Table 1. Comparison with existing interleaved multimodal benchmarks. Inter. Comb.: Interleaved combinations of modalities. Cap. per Instance: Capability per instance. Difficulty Tax.: Difficulty taxonomy.

Benchmarks	Domains	Num.	Inter. Comb.	Cap. per Instance	Eval. Metric	Difficulty Tax.	Any-to	to-Any	Modalities
ITLVD-BENCH [34]	10	815	2	Single	5	✗	✗	✗	📝 🏞
OpenING [79]	8	5,400	4	Single	7	✗	✗	✗	📝 🏞
ISG-Bench [6]	8	1,150	3	Single	4	✗	✗	✗	📝 🏞
CoMM [8]	3	/	4	Single	3	✗	✗	✗	📝 🏞
MMIE [67]	10	20,103	3	Single	7	✗	✗	✗	📝 🏞
UNIM (Ours)	**30**	**31,026**	**41**	**Multiple**	**13**	✓	✓	✓	📝 🏞 🔊 🎬 📁 💻 🧊



UniM Evaluation Suite



Evaluation Dimensions and Metrics

3 dimensions, **13** metrics

(1) Semantic Correctness & Generation Quality: *SC*, *GQ*, $SQCS^{abs}$, $SQCS^{rel}$.

(2) Response Structure Integrity: StS^{abs}, StS^{rel}, LeS^{abs}, LeS^{rel}.

(3) Interleaved Coherence: *HC*, *SH*, ICS^{abs}, ICS^{rel}.

Supporting Rate: τ.

UniMA: A Unified A2A Agentic Model

Core: Traceable Evidence Reasoning Module

(1) Structured Evidence Reasoning Chain: reason and generate a final report for guiding the downstream any modality generation.

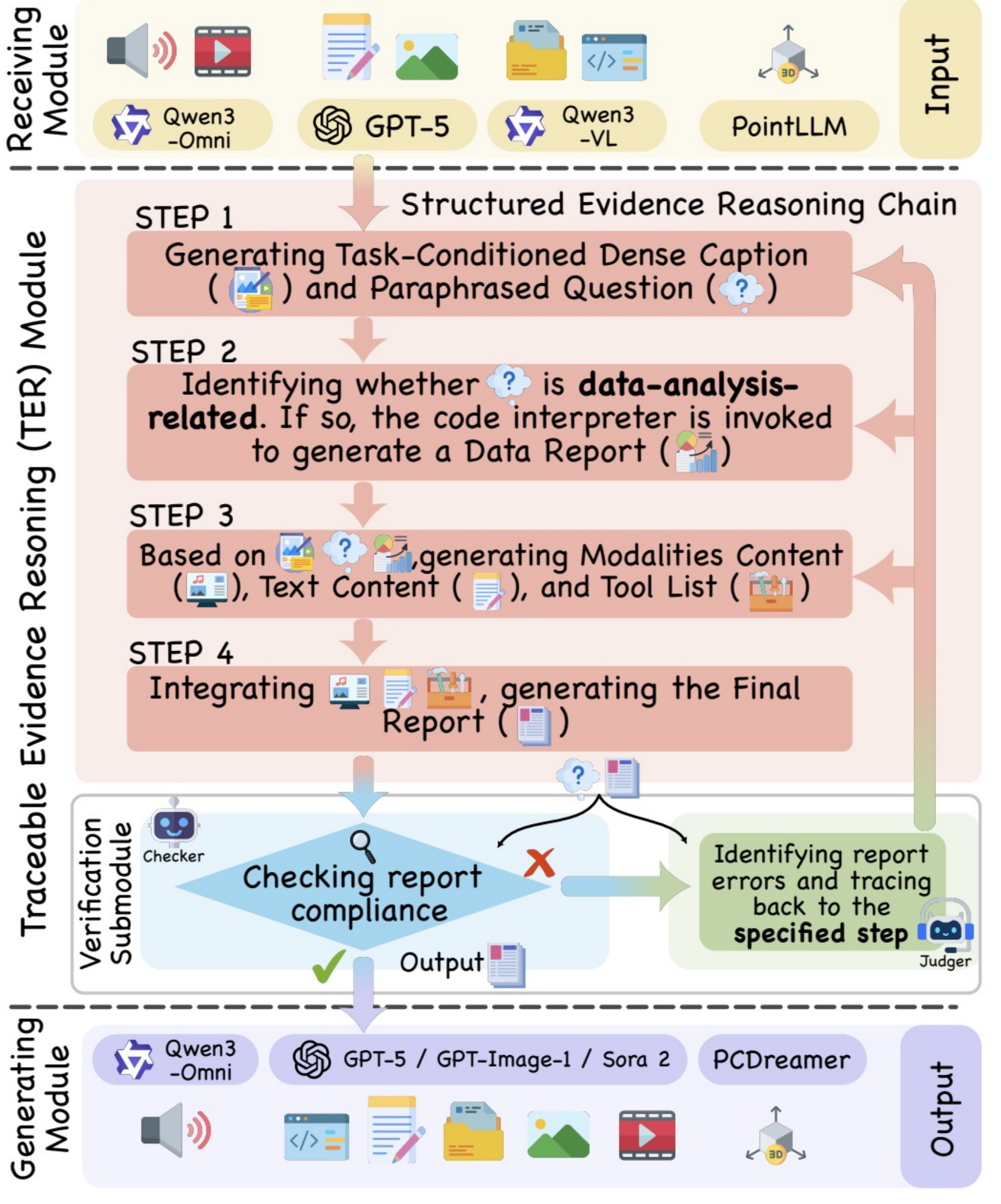
(2) Verification Submodule: Make sure that final report is both correct and aligned with the user's original question as the guidance.

Model	Multi-I Input	Multi-I Output	Any-M Input	Any-M Output	Any-I Input	Any-I Output
AnyGPT	✗	✗	✗	✗	✗	✗
NExT-GPT	✗	✗	✗	✗	✗	✗
MIO	✓	✓	✗	✗	✗	✗
UNIMA	✓	✓	✓	✓	✓	✓

Multi-I Input / Output: multiple items of the same modalities in the input / output.

Any-M Input / Output: any modality combinations in the input / output.

Any-I Input / Output: any number of items of the same modality in the input / output.


Receiving Module
Qwen3-Omni
GPT-5
Qwen3-VL
PointLLM
Input
Traceable Evidence Resoning (TER) Module
Structured Evidence Reasoning Chain
STEP 1
Generating Task-Conditioned Dense Caption ( ) and Paraphrased Question (?)
STEP 2
Identifying whether ? is data-analysis-related. If so, the code interpreter is invoked to generate a Data Report ( )
STEP 3
Based on ?, generating Modalities Content ( ), Text Content ( ), and Tool List ( )
STEP 4
Integrating , generating the Final Report ( )
Verification Submodule
Checker
Checking report compliance
Output
Identifying report errors and tracing back to the specified step
Judger
Generating Module
Qwen3-Omni
GPT-5 / GPT-Image-1 / Sora 2
PCDreamer
Output

Experiments Results

(1) Main Results

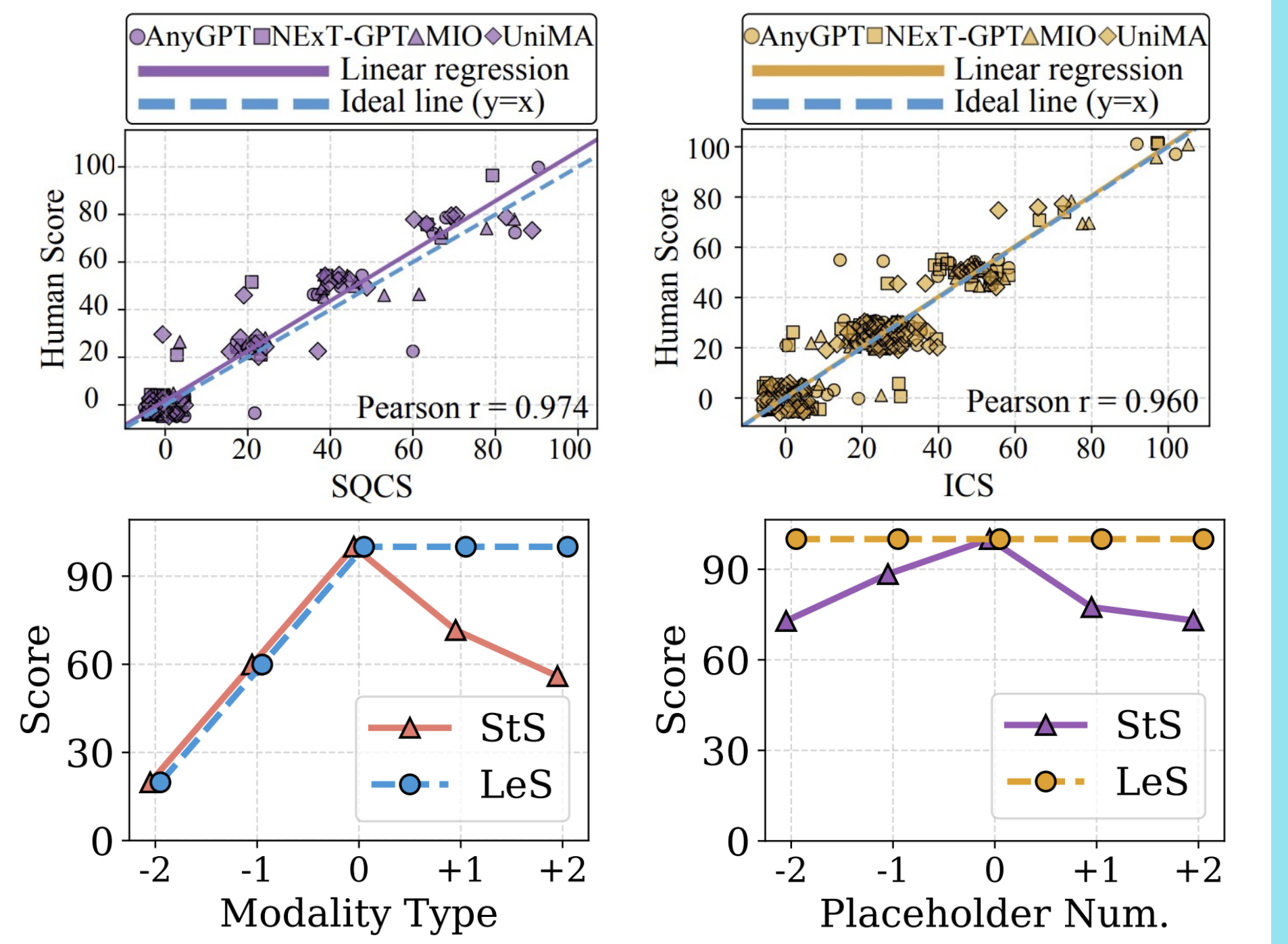
Model	Natural Science StS^{abs}	Natural Science LeS^{abs}	Natural Science StS^{rel}	Natural Science LeS^{rel}	Social Science StS^{abs}	Social Science LeS^{abs}	Social Science StS^{rel}	Social Science LeS^{rel}	General Area StS^{abs}	General Area LeS^{abs}	General Area StS^{rel}	General Area LeS^{rel}
AnyGPT [76]	12.9	27.8	12.2	21.4	14.9	16.6	14.5	16.2	12.5	16.4	9.8	13.6
NExT-GPT [64]	2.0	2.2	1.2	1.3	1.3	1.7	1.2	1.5	2.2	2.5	1.4	1.7
MIO [59]	1.3	1.9	0.9	1.3	4.1	5.2	4.0	5.1	3.3	3.8	2.4	2.9
UNIMA	50.8	62.2	50.8	62.2	58.1	72.9	58.1	72.9	71.3	84.3	71.3	84.3

Field	Models	SC	GQ	$SQCS^{abs}$	τ	$SQCS^{rel}$
Natural Science	AnyGPT [76]	13.7	37.9	11.1	90.4	10.7
	NExT-GPT [64]	8.4	23.4	6.2	62.0	2.9
	MIO [59]	19.7	29.1	15.9	59.2	10.0
	UNIMA	59.8	79.7	57.3	100	57.3
Social Science	AnyGPT [76]	18.0	23.8	15.5	94.7	14.7
	NExT-GPT [64]	16.8	31.9	13.3	89.0	10.8
	MIO [59]	25.2	32.8	21.4	80.8	16.1
	UNIMA	76.2	81.0	72.7	100	72.7
General Area	AnyGPT [76]	19.0	30.1	17.9	90.3	17.2
	NExT-GPT [64]	5.4	30.0	4.4	76.0	3.4
	MIO [59]	24.8	37.5	21.2	71.7	15.2
	UNIMA	64.7	83.6	62.2	100	62.2

Field	Models	HC	SH	ICS^{abs}	ICS^{rel}
Natural Science	AnyGPT [76]	39.9	46.3	41.8	38.5
	NExT-GPT [64]	23.5	26.1	24.9	16.3
	MIO [59]	49.4	63.7	52.1	31.8
	UNIMA	68.4	71.9	69.1	69.1
Social Science	AnyGPT [76]	31.3	35.3	32.1	29.2
	NExT-GPT [64]	24.5	27.1	21.4	19.0
	MIO [59]	46.3	55.0	51.6	42.0
	UNIMA	73.1	76.5	73.8	73.8
General Area	AnyGPT [76]	36.5	41.9	43.6	31.3
	NExT-GPT [64]	27.9	31.1	28.1	20.0
	MIO [59]	68.3	77.7	60.0	45.7
	UNIMA	68.7	74.3	69.8	69.8

	SQCS	ICS	StS	LeS
UNIMA	**85.1**	**63.4**	**52.7**	**82.6**
w/o **TER**	72.9 (-12.2)	56.6 (-6.8)	16.4 (-36.3)	21.8 (-60.8)
w/o **TCDC**	78.4 (-6.7)	57.7 (-5.7)	46.2 (-6.5)	82.1 (-0.5)
w/o **Ver.**	72.9 (-12.2)	54.7 (-8.7)	38.3 (-14.4)	66.8 (-15.8)

(2) Rationality Verification



Analysis and Observations

➤ Baseline models exhibit poor performance of absolute metrics on UniM.

➤ Notably lower relative scores owing to baselines' limited support for diverse interleaved input modality.

➤ Model performance varies significantly across different fields and domains.

➤ UniMA shows clearly superior performance on UniM, serving as a reasonable baseline.

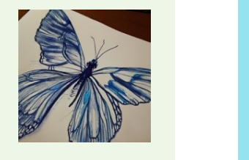
(3) Case Study

▶ **User Query:** Watch the instructional video in *<video1>* , please provide a text-image tutorial about how to draw a blue morpho butterfly using watercolor techniques.

<video1>

▶ **NExT-GPT Response:** Certainly! I'll be happy to help you with that. In this audio, I'll guide you through the process of drawing a blue butterfly using watercolor techniques. First, we'll start by discussing the basic techniques for creating a blue butterfly, including how to hold the pencil and how to get the right shade of blue. Then, we'll explore different ways to add details like highlights and shading to your butterfly. By the end of the video, you'll have a beautiful blue butterfly that you can be proud of! Enjoy the process! **(No image generation)**

▶ **MIO Response:** A person first sketches the outline of a butterfly with a pen, then colors it, and finally finishes drawing a blue butterfly. *<image1>* **(Missing steps)**


<image1>

▶ **UniMA Response:** Get materials ready — watercolor paper, round brushes, rigger, blue palette, waterproof liner, white gouache, water, towels… *<image1>* Lightly sketch the butterfly; mark wings/veins; optional masking fluid… *<image2>* Mix a range of blues → deep ultramarine… *<image3>* Wet one wing and lay the lightest blue wash… *<image4>* Add loose background splatters for expressive effect… *<image5>* Lift small highlights; add minimal white gouache for shimmer… *<image6>* Add darker edges + small eye-spots; soften inward… *<image7>* Remove masking; re-ink if needed; add final highlights — your butterfly is complete… *<image8>*

<image1><image2><image3><image4><image5><image6><image7><image8>